

Lecture 3

The Anatomy of LM386

Prof Peter YK Cheung
Imperial College London

URL: www.ee.ic.ac.uk/pcheung/teaching/EE2_CAS/
E-mail: p.cheung@imperial.ac.uk

In this week's lecture, I will "dissect" the LM386 audio amplifier. The purpose of this is to expose you to what a real-life amplifier looks like inside with the packaging removed.

In Year 1, you have already learned how different types of transistor circuits work: common-emitter amplifier, differential amplifier, emitter follower etc.. You have also used SPICE to simulate, at transistor level, a typical operational amplifier as part of the Laboratory coursework.

In this lecture, we will examine in detail how the theory you explored last year is found in one of the most popular audio amplifiers used in industry. The LM386 was designed by National Semiconductors in the '70s and has remained a favourite among audio enthusiasts due to its low cost, low voltage single supply operation, minimum number of external components and low power consumption. It is particularly suitable for battery operate, portable designs.

The amplifier is built with BJTs, not FETs or MOSFETs. It is important to remember that although the LM386 contains an op-amp inside, it is NOT itself an op-amp, and it cannot be used as such.

What you should know already?

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C-E Amplifier Revisited Quiescent Analysis

- KVL on input side

$$V_{BIAS} = I_B R_B + V_{BE}$$

and $V_{BE} \approx 0.7\text{ V}$, $I_E = I_B + I_C$

$$\Rightarrow I_E \approx \frac{V_{BIAS} - V_{BE}}{R_E}$$

- As before:

$$V_E = I_E R_E$$

BJT Current Mirror

INPUT SIDE

- For transistors identical apart from area ratio A :

$$I \approx I_{REF} \frac{A}{A_{REF}}$$

where we have used $I_S \propto A$, and

The Differential Pair – large signal

- If transistors are matched, then:

$$I_{C1} = \frac{I}{1 + \exp(-V_D/V_T)} \quad I_{C2} = \frac{I}{1 + \exp(V_D/V_T)}$$

$V_D = (V_{IN1} - V_{IN2})$ is the DIFFERENTIAL INPUT VOLTAGE

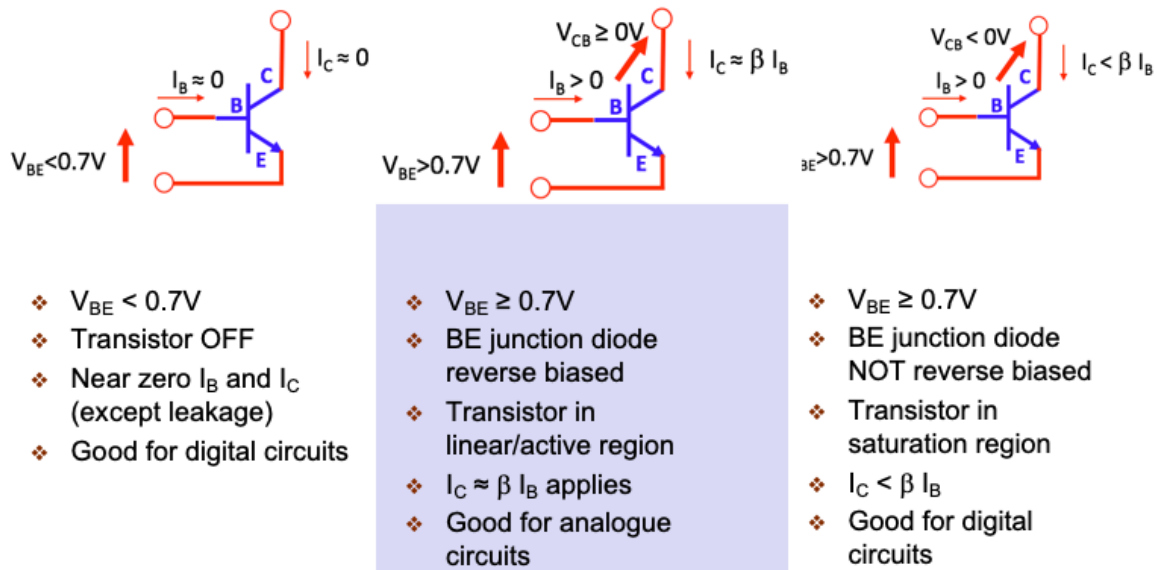
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Here are extracts of some of the slides you used in the Analysis and Design of Circuits module in Year 1, where you have learned all the equations relating to currents and voltages in different configurations of bipolar transistor circuits.

While knowing equations governing the working of circuits is important, and able to analyse circuits in a rigorous manner is essential for an electronic engineer, it is also important for you to acquire some intuition about circuits. Such intuitions would allow you to have deep insights on why you might consider adding a transistor at a certain place may be worth exploring. This ability would allow you to “see” how a circuit works and understand why it has limitations. This helps you inventing new circuits that work better than previous designs.

The approach taken in this lecture would therefore be to minimize equations and to provide you with some “rules of thumb” that encourage such intuition.

Rules of thumb – Bipolar Junction Transistor (BJT)



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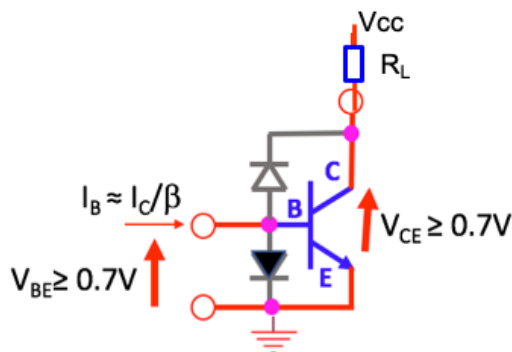
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Here are some basic rules relating to bipolar transistors – NPN in this case. You should be familiar with these from Year 1, but here it is again in a condensed form.

A NPN transistor has three normal operating states:

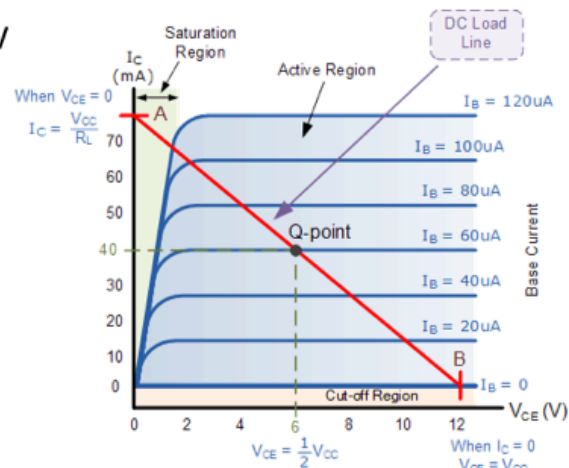
- 1. Off state** - when base-emitter voltage is lower than 0.6 – 0.7V, the base-emitter junction diode is not forward biased. The transistor is not conducting. There is nearly no current through any of its three terminals (except leakage currents). This mode of transistor is found operating in digital circuits as an OFF switch.
- 2. Linear or Active state** – if the base-emitter junction diode is forward biased because $V_{BE} \geq 0.7V$ (could be lower), the transistor is conducting. Furthermore, the base-collector junction diode **must be reverse biased**. The ratio of collector current to base current is a constant (β). The transistor is operating in the linear and active region. Transistors in this state is usually found in good quality analogue circuits that are working properly.
- 3. On state** – if both the base-emitter and base-collector junction diodes are forward biased, the collector-emitter voltage drops below, say, 0.3V, the transistor is in saturation mode. The collector to base current is no longer governed by β ($I_C/I_B < \beta$). The transistor behaviours like an ON switch in digital circuits.

NPN Transistor in Linear Region



- ❖ BE junction diode forward biased (ON)
- ❖ BC junction diode OFF
- ❖ I_C is often determined by DC bias circuit
- ❖ I_B then fixed by β
- ❖ V_{CE} can take on any value between 0.3V and V_{CC}

- ❖ Quiescent (DC) condition – biasing
- ❖ Ensures transistor in linear region
- ❖ Operating condition determined by I_C
- ❖ Use large signal model



Since we are considering analogue audio amplifiers, we are interested in transistors working in the **linear** or **active** region.

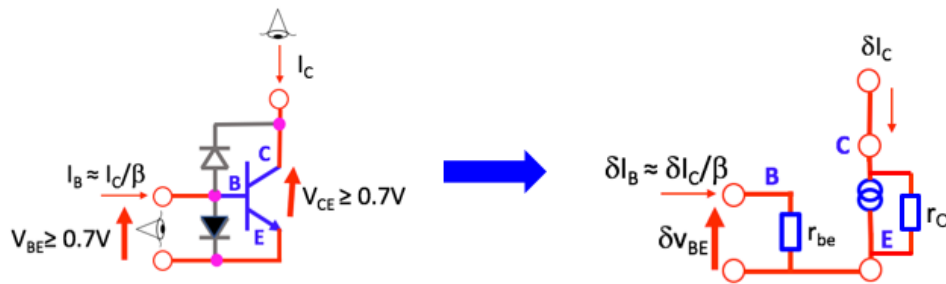
Remember again, looking into the base terminal of an NPN transistor, you see two diodes: a base-emitter diode, a base-collector diode, as shown in the slide. In this mode, the BE diode is forward biased; the BC diode is reverse biased.

To ensure that a NPN transistor is operating in the active region, one must construct circuits around the transistor such that when no signal is applied (**quiescent condition**), the transistor is operating in the required conditions. Before a transistor in an analogue circuit can do its thing (e.g. amplify signals), one must bias the transistor properly and establish all voltages and currents under DC condition (i.e. no signal).

Shown on the slide is a typical I_C vs V_{CE} characteristics of an NPN transistor. A common way to establish bias is to add a load resistor between collector and V_{CC} . This establishes what is called a "**load line**". The idea is to bias the circuit so that under quiescent condition, the transistor "sits" at the middle of the region, which in this case is $V_{CE} = V_{CC}/2$. In this way, the output voltage at the collector will have maximum voltage swing (or range).

To ensure that the base-collector junction diode remains non-conducting, V_{CE} must not drop below 0.3V.

Small signal behaviour of NPN transistor



- ❖ Quiescent I_C determine operating point
- ❖ Small signal model – 2 resistors r_{be} and r_o , and a current source
- ❖ r_o large - can be omitted

- ❖ $\delta i_B = \delta v_{BE} / r_{BE}$
- ❖ $\delta i_C = g_m \delta v_{BE}$
- ❖ $\delta i_E = \delta i_B + \delta i_C$

- ❖ current gain: $\beta = \delta i_C / \delta i_B$
- ❖ transconductance: $g_m = \delta i_C / \delta v_{BE} = I_C / V_T$
- ❖ Input resistance: $r_{be} = \delta v_{BE} / \delta i_B = \beta V_T / I_C$

Once a transistor is properly biased at DC, we can consider the operation of the transistor when we inject a signal. For this, we use a simplified small signal model.

For a BJT transistor, the small signal model have parameters that are governed by the operating point of the BJT (i.e. the bias condition), particularly the value of I_C .

The small signal consists of an input resistance r_{be} as viewed from the base-emitter port, a voltage controlled current source between base and emitter, and an output resistance r_o in parallel with the current source between collector and emitter. Usually r_o is large enough to be removed from the model.

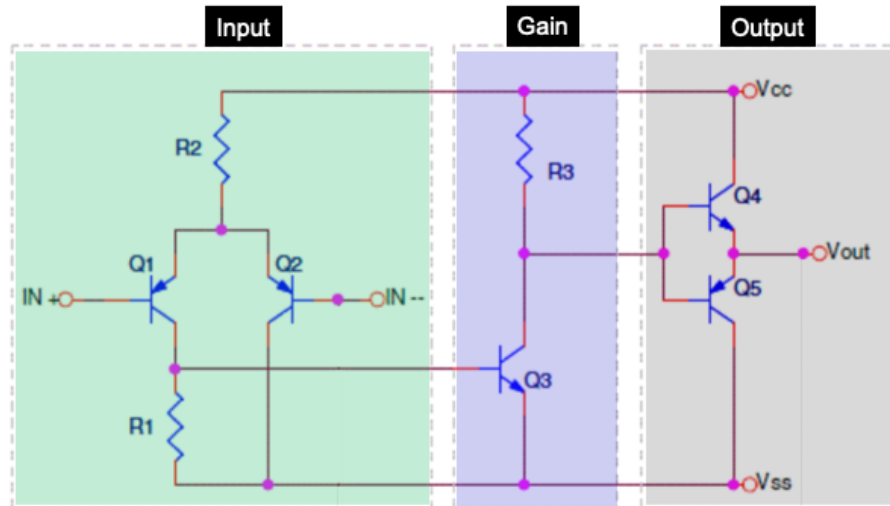
The three most important parameters for a NPN BJT are as shown:

1. **Current gain** β - typically 20 to 200, determined by the fabrication process;
2. **Transconductance** g_m – depends on collector bias current I_C ;
3. **Input resistance** r_{be} – depends on both collector bias and property of the transistor.

Inside a typical op-amp

❖ Three stages architecture:

1. Differential input stage – long-tail pair (Yr 1 Circuits part 2, adc_9, slides 8-14)
2. Voltage gain stage – common emitter amp (adc_6, slides 3-7)
3. Output drive stage – push-pull circuit



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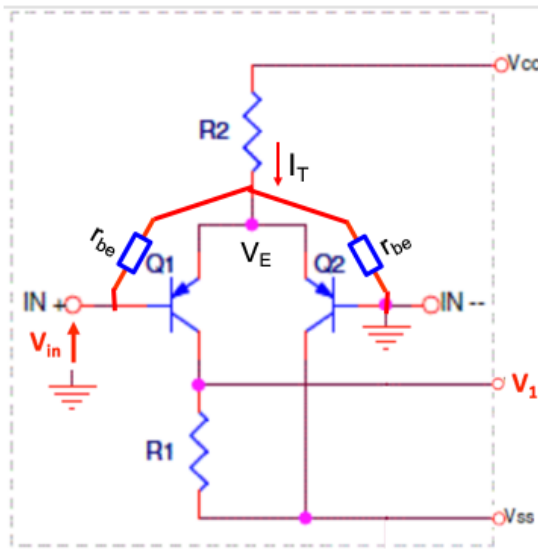
Let us now turn our attention to a generic op-amp architecture. You have already encountered this in Year 1 Spring Term Lab ADC_4, when you simulated an op-amp at transistor level using SPICE.

A typical op-amp consists of three stages:

- 1) The input stage provides differential inputs IN+ and IN- and gives the op-amp the ability to reject common-mode signals. (Common-mode signal is a signal that is applied to both IN+ and IN- simultaneously.)
- 2) The second stage is a common-emitter gain stage that provides most of the gain in an op-amp.
- 3) The third stage is the output stage that provides the output current capability.

In the following slides, we will examine how each stage works, and derives the gain equation from an intuitive perspective.

Differential Input Stage – differential gain



$$A_{V_{diff}} = \delta v_1 / \delta v_{in} = -\frac{1}{2} g_m * R_1$$

Consider $V_{in} = 0$, $IN -$ is grounded

- ❖ $V_E \approx 0.7V$
- ❖ $I_T \approx (V_{CC} - 0.7) / R_2$
- ❖ $I_1 \approx I_2 \approx \frac{1}{2} I_T \approx \frac{1}{2} (V_{CC} - 0.7) / R_2$
- ❖ $V_1 = R_1 * I_{C1} + V_{SS}$

Apply small input signal δV_{in}

- ❖ $\delta v_E \approx \frac{1}{2} \delta v_{in}$
- ❖ $\delta i_T \approx \frac{1}{2} \delta v_{in} / R_2 \approx 0$
- ❖ $\delta i_1 \approx -\delta i_2 = -g_m * \frac{1}{2} \delta v_{in}$
- ❖ $\delta v_1 = R_1 * \delta i_1 = -R_1 * g_m * \frac{1}{2} \delta v_{in}$

This circuit is called a “long-tail pair”, and it consists of two transistors Q1 and Q2, with a tail resistor R2, and load resistor R1.

Let us first consider the quiescent (DC) condition when both $IN+$ and $IN-$ are connected to ground.

1. V_E , the emitter voltage of Q1 and Q2, is around 0.7V above ground. This determines the tail current I_T and establishes bias condition for Q1 and Q2.
2. Since Q1 and Q2 matched, their currents MUST be identical. Therefore $I_1 = I_2 = \frac{1}{2} I_T$.
3. R1 can then be chosen to determine the bias voltage of the output V_1 .

For differential input, assuming that signal is small at $IN+$ and is δV_{in} .

Viewing from $IN+$ terminal, the input looks like two resistors r_{be} in series to ground. This works as a voltage divider such that

$$\delta V_{BE2} = -\delta V_{BE1} = \delta V_E = \frac{1}{2} \delta V_{in}$$

Since $\delta V_E \ll V_E$ because V_E is fixed by V_{BE2} , I_T is essentially unchanged (i.e. $\delta I_T \approx 0$).

The change in Q1 current, δI_{C1} , is determined by its transconductance:

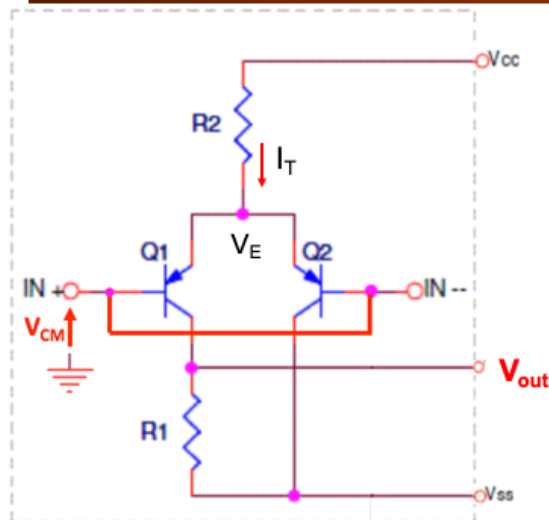
$$\delta I_1 \approx -\delta I_2 = -g_m * \frac{1}{2} \delta v_{in}$$

Hence, we can calculate the differential voltage gain from V_{in} to V_1 as:

$$A_{V_{diff}} = -\frac{1}{2} g_m * R_1$$

To increase differential gain, we have to either increase g_m (by reducing R2, thus increasing I_{C1}) or R1.

Differential Input Stage – common mode gain



Apply v_{CM} to both inputs

- ❖ $\delta v_E \approx v_{CM}$
- ❖ $\delta i_T \approx -\delta v_E / R_2 = -v_{CM} / R_2$
- ❖ $\delta i_{C1} \approx \delta i_{C2} = \frac{1}{2} \delta i_T = -\frac{1}{2} v_{CM} / R_2$
- ❖ $\delta v_{out} \approx -R_1 / 2R_2 * v_{CM}$

$$A_{V_{cm}} = \delta v_{out} / v_{CM} = -R_1 / 2R_2$$

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A good differential amplifier should amplify differential signals WITHOUT amplifying common-mode signal. Therefore we need to consider the common-mode gain of the long-tail pair circuit. For this, we connect IN+ and IN- together to a common voltage V_{CM} .

The input signal V_{CM} is transfer directly to the emitter of Q1, Q2 since V_{BE} of Q1 and Q2 remain more or less fixed at 0.7V. Therefore the tail current changes by:

$$\delta i_T \approx -\delta v_E / R_2 = -v_{CM} / R_2$$

Since both Q1 and Q2 have the same V_{BE} and they are matched, their collector currents must be identical. Hence

$$\delta i_{C1} \approx \delta i_{C2} = \frac{1}{2} \delta i_T = -\frac{1}{2} v_{CM} / R_2$$

Therefore the change in the output voltage is simply:

$$\delta v_{out} \approx -R_1 / 2R_2 * v_{CM}$$

From this we calculate the common-mode gain to be:

$$A_{V_{cm}} \approx \delta v_{out} / v_{CM} = -R_1 / 2R_2 \quad \text{and}$$

$$A_{V_{diff}} \approx -\frac{1}{2} g_m * R_1$$

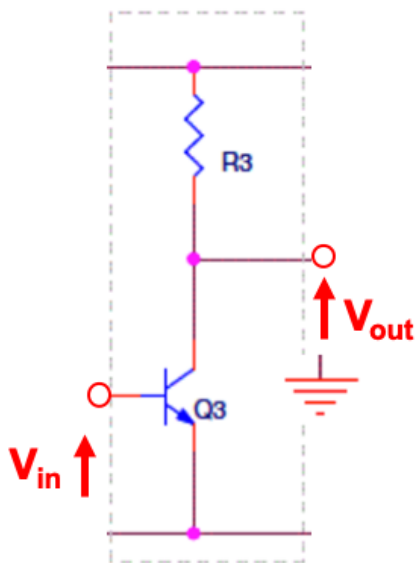
Common-Mode Rejection Ratio = Differential Gain/Common-mode Gain

$$CMRR = A_{V_{diff}} / A_{V_{cm}} \approx g_m * R_2$$

Here we illustrate a contradiction: for larger CMRR, R_2 needs to be increased.

However, for large differential gain, we want to increase g_m and I_{C1} , which means R_2 should be reduced. This contradiction limits the differential gain of the input stage.

Gain stage



- ❖ Common Emitter amplifier (Yr 1st ADC part 2 Lecture 6, S7)
- ❖ $\delta v_{out} = -g_m \delta v_{in} * R_3$
- ❖ $A_V = \delta v_{out} / \delta v_{in} = -g_m R_3$
- ❖ Increase g_m and R_3
- ❖ Most of the voltage gain produced by this stage

Due to the contradictory requirements of differential and common-mode gains, op-amp uses a second stage of amplification to provide high open-loop voltage gain. The second stage is a simple common-emitter stage with a single transistor. The gain of this stage is straight forward.

The input signal is applied directly to the base-emitter junction:

$$\delta V_{be} = \delta V_{in}$$

The NPN transistor "translates" this base-emitter voltage change to a change in collector current determined by the transconductance parameter g_m .

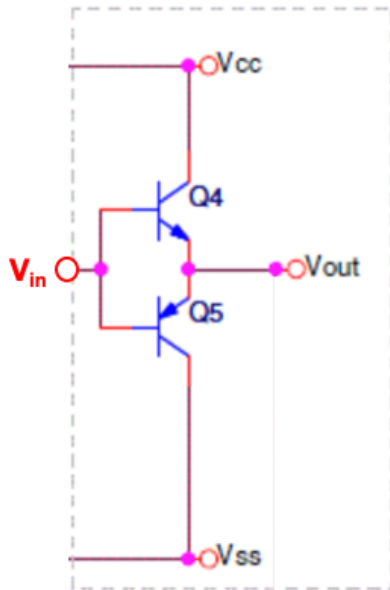
The overall voltage gain of this stage is therefore:

$$A_V = \delta v_{out} / \delta v_{in} = -g_m R_3$$

To increase stage 2 gain, one can increase g_m by increasing the bias current to Q3, and/or increase the load resistance R_3 .

Unlike the input differential stage, this stage has no impact on CMRR and there is no restriction (at least in theory) to the gain of this stage. As a result, most of the voltage gain is obtained from this 2nd stage.

Output Stage



- ❖ Yr 1st ADC part 2 Lecture 6, S3-5
- ❖ Q4 is emitter follower (Common-Emitter) for sourcing current to Vout (PUSH)
- ❖ Q5 is another emitter follower for sinking current from Vout (PULL)
- ❖ This is known as a PUSH-PULL circuit
- ❖ $\delta V_{out} \approx \delta V_{in}$

The third stage consists of complementary transistors Q4 and Q5. They each acts as an emitter follower circuit for the positive and negative half of V_{in} (as a sine wave) relative to the quiescent condition of V_{out} . This circuit structure is known as the “Push-Pull” configuration or a class AB amplifier.

If $V_{in} \geq V_{out}(\text{quiescent}) + 0.7V$, Q4 provides current to drive V_{out} to $V_{in} - V_{BE4}$. This is known as “push” action - it sources current to the output load.

If $V_{in} \leq V_{out}(\text{quiescent}) - 0.7V$, Q5 absorbs current to drive V_{out} to $V_{in} + V_{BE5}$. This is known as “pull” action - it sinks current from the output load.

This circuit incurs significant distortion when $-0.7 \leq V_{in} \leq +0.7$. During such crossover region, neither Q4 nor Q5 is conducting and the output is not effectively driven.

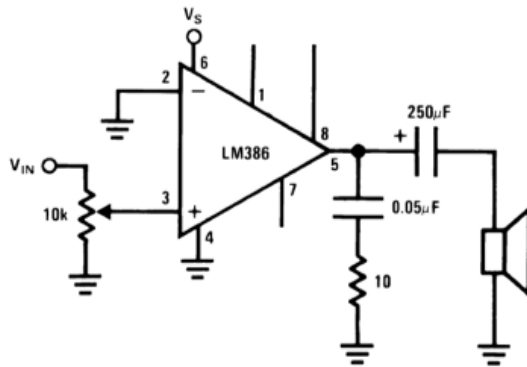
The small signal voltage gain of the 3rd stage is approximately 1.

Driving 8Ω speaker with LM386

LM386

Low Voltage Audio Power Amplifier

Amplifier with Gain = 20 Minimum Parts



Features

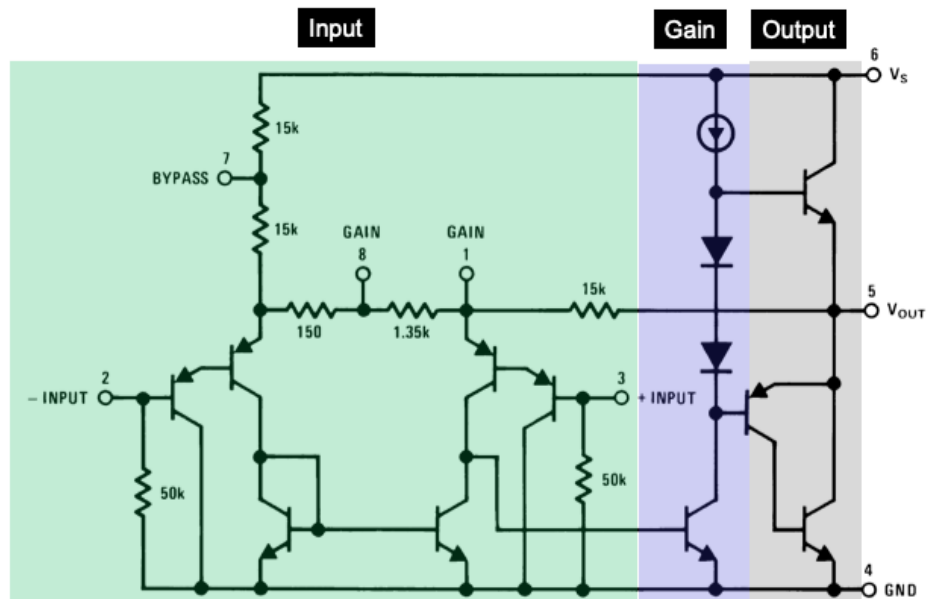
- Battery operation
- Minimum external parts
- Wide supply voltage range: 4V–12V or 5V–18V
- Low quiescent current drain: 4mA
- Voltage gains from 20 to 200
- Ground referenced input
- Self-centering output quiescent voltage
- Low distortion: 0.2% ($A_V = 20$, $V_S = 6V$, $R_L = 8\Omega$, $P_O = 125mW$, $f = 1kHz$)
- Available in 8 pin MSOP package

Now we are ready to examine exactly how LM386 works by studying the detail circuit inside the package.

This amplifier is NOT an op-amp because it cannot be connected in the usual way with an op-amp as you did with the MCP6002 from the previous lecture. Instead, this circuit contains its own internal feedback network such that it has an inherent voltage gain of around 20 (or 26dB).

Shown here is the circuit from the datasheet showing the basic configuration of the LM386 as a x20 amplifier driving a speaker. It also provides a summary of its desirable features. Most notable are the first two: battery operation, meaning that it works with a single power supply, and minimum external parts.

LM386 internal circuit

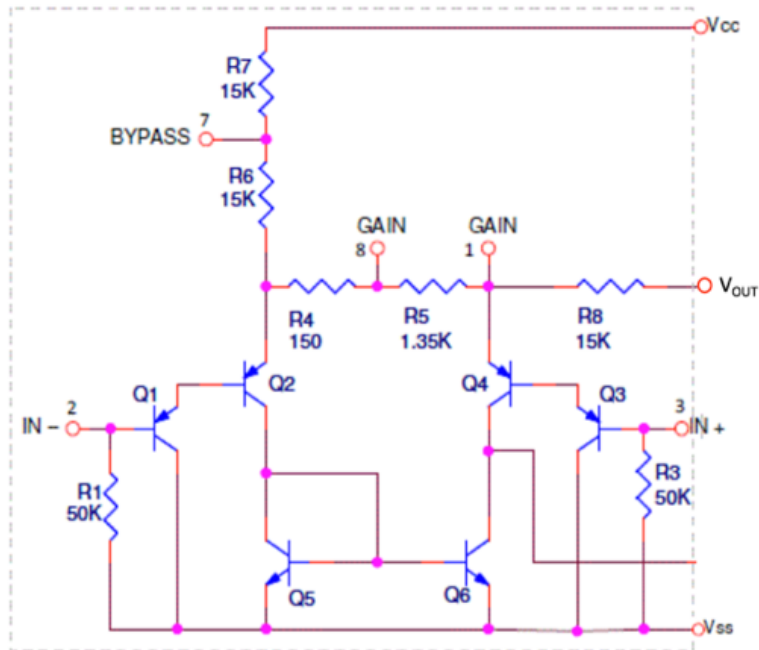


This shows the full internal circuit of LM386 at transistor level (except the current source in the 2nd stage).

This follows the classic three stage op-amp architecture with several refinements.

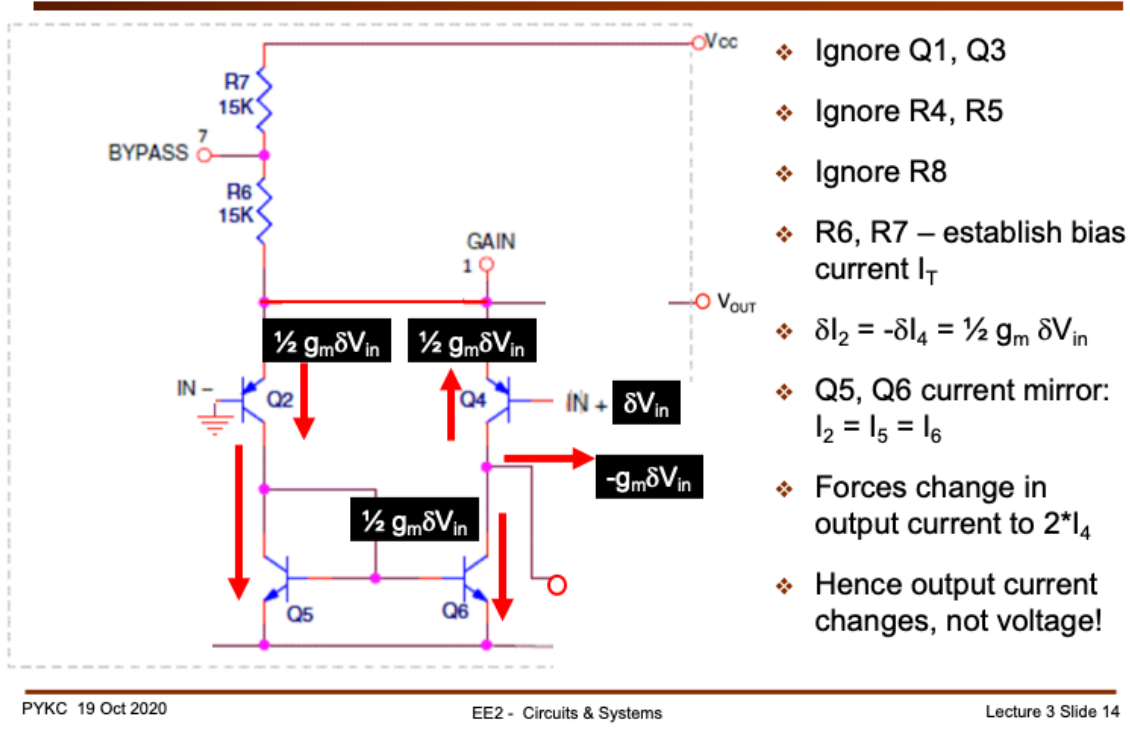
We will now consider each stage separately and discuss the design choices made and reasons why.

LM386 Input Stage (1)



The input stage is a long-tail pair with additional input transistors Q1, Q3 and R1, R3. Here all components are labelled and numbered for ease of reference in the explanation. Otherwise, it is identical to the circuit you find in the datasheet.

LM386 Input Stage (2)



Let us start by simplifying the circuit and ignore Q1 and Q3, the additional input stages. Further, let us remove R8, the feedback resistor from V_{out} . Since R4 and R5 also form part of the feedback network and is relatively small compared with other resistor values, let us assume them to be zero for now (i.e. short circuit).

Here we can see that this is a conventional long-tail pair except that the load resistor at the collector of Q4 is replaced by a circuit Q5 and Q6.

Consider $IN+$ goes up by δV_{in} . This increases V_{BE2} by $\frac{1}{2} \delta V_{in}$ and decreases V_{BE4} by $\frac{1}{2} \delta V_{in}$. Therefore $\delta I_4 = -\delta I_2 = -\frac{1}{2} g_m \delta V_{in}$.

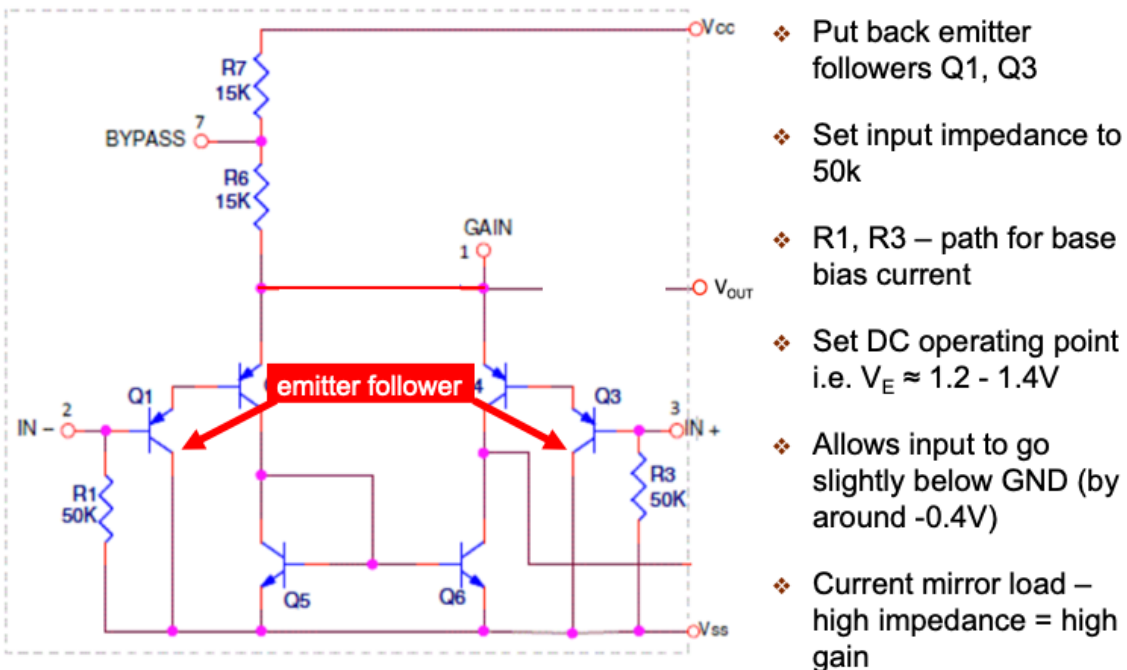
However, Q2 and Q5 are in series and share the same current, i.e. $\delta I_5 = \delta I_2$. Therefore δI_2 is mirror by Q6 onto the right branch of differential pair as $\delta I_6 = \delta I_2$.

Since $\delta I_4 = -\delta I_2 = -\frac{1}{2} g_m \delta V_{in}$, the change in output current is

$$\delta I_{out} = -\delta I_2 + \delta I_4 = -2 \delta I_2 = -g_m \delta V_{in}$$

This design produces an output current gain, not voltage gain!

LM386 Input Stage (3)



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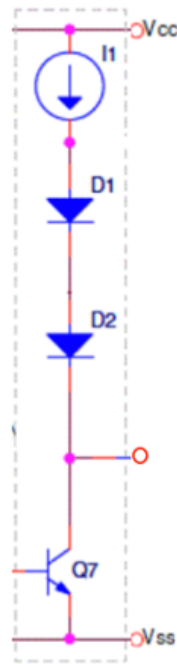
The long-tail pair inputs are “buffered” with two emitter follower type transistor Q1 and Q3. This additional circuit serves several purposes:

1. R1 and R2 fix the input resistances at 50k ohms and provide a DC path for the base current to follow, biasing the input stage. It avoids the inputs of the amplifier become floating, and therefore the amplifier output is also biased properly even the input terminals are open circuit.
2. It allows input to swing below V_{SS} (i.e. negative) and ensure that all transistors stay in the active region. Assuming that V_{SS} is GND, this allows inputs to swing between $\pm 0.4V$.

Why is this the case? Let us assume $IN+$ is $-0.4V$, which is negative relative to V_{SS} . Since Q3 and Q5 are PNP transistors, their base-emitter junction diodes will be forward biased. For Q3 to work in the active region and thus behave like an emitter follower, the base-collector junction needs to remain reverse biased. Fortunately $-0.4V$ is not sufficient to turn on the base-collector diode for Q3.

So far we have ignored R4, R5 and R8 because they are part of the feedback network, and they do not belong to the differential stage. We will come back to them later.

LM386 Gain Stage



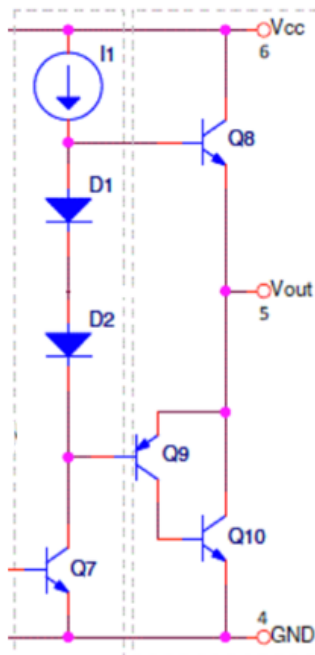
- ❖ Common emitter amplifier with active load – constant current source I_1
- ❖ Q7 g_m determined by I_1
- ❖ D1, D2 are for output stage - 1.2V constant offset
- ❖ D1, D2 has no impact on gain of 2nd stage
- ❖ Q7 collector loads a constant current source
- ❖ Gain is not through g_m , but through β
- ❖ $\delta I_{out} = \beta_7 \delta I_{B7}$

Unlike the classical 2nd voltage gain stage, the 2nd stage of the LM386 has a collector **current source** load instead of a **resistive load**. This is convenient because as seen in earlier slides, the differential stage provides an amplified current output, which feeds the base of Q7.

Since the load of the 2nd stage is the current source I_1 , the output of this stage is again a current, $\delta I_{out} = \beta_7 \delta I_{B7}$.

The output current feeds the output stage.

LM386 Output Stage



- ❖ Q8 push emitter follower sourcing current
- ❖ Q9, Q10 pull emitter follower sinking current
- ❖ D1, D2 forward bias due to I_1
- ❖ Keep Q8 and Q9 in linear region – reduce distortion
- ❖ Q9 PNP has poor current gain
- ❖ Combine with Q10 NPN to boost current gain
- ❖ Q9, Q10 - called a “Sziklai” pair (different from Darlington pair)
- ❖ $A_v \approx 1$

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This output stage is very similar to what has been discussed before except for two features.

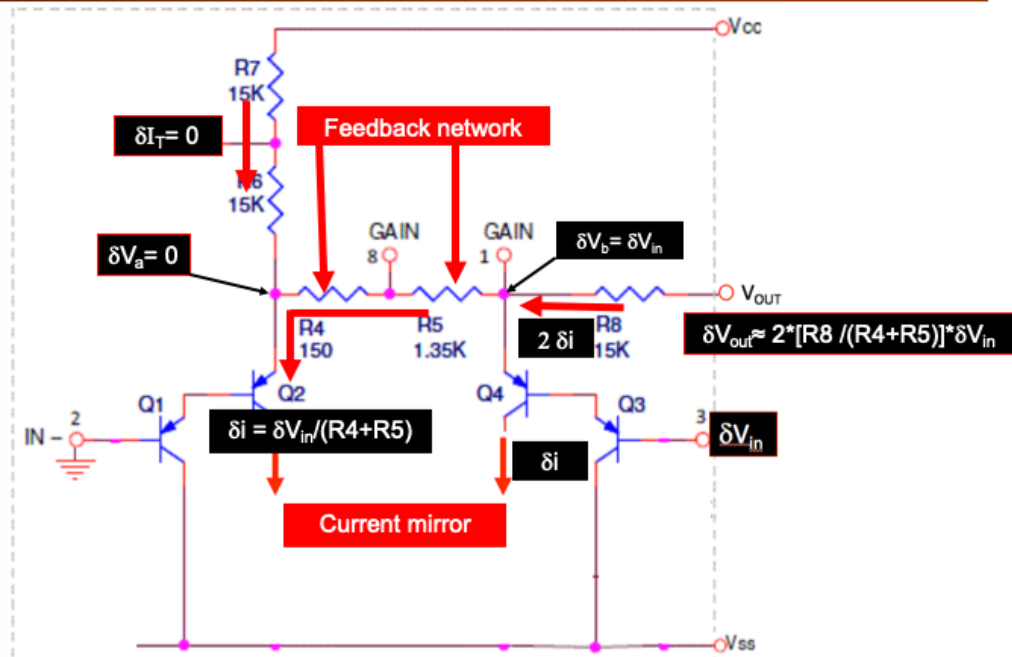
As discussed earlier in slide 10, the simple push-pull circuit exhibits severe distortion because the NPN and PNP transistors are not conducting whenever their base-emitter voltages are below 0.7V. This is mitigated by adding two diodes D1 and D2 as shown. These two diodes are forward biased via the constant current source I_1 , and they ensure that both Q8 and Q9 are always operating in the linear region (i.e. base-emitter forward biased). This eliminates much of the distortion.

The second feature is that the PNP transistor is enhanced with Q10, a NPN transistor. The reason is that PNP transistors typically have significantly lower current gain than NPN transistors.

By adding Q10, the combined current gain is boosted to: $\beta \approx \beta_9 \beta_{10}$. This configuration is known as a “**Sziklai** transistor pair”. It is different from **Darlington pair** which consists of two transistors of the SAME TYPE connected in cascade.

Sziklai transistor pair, like a Darlington pair, provides the same improved current gain (product of the two transistors' current gains), but it requires only one V_{BE} to turn on, and not two as in a standard Darlington configuration.

LM386 Gain is $2 \cdot R8 / (R4 + R5) \approx 20$



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Now let us put back the feedback resistors R4, R5 and R8, and establish why with these resistors in the circuit, LM386 provides a gain of approximately 20.

Here is a simplified version of the entire circuit schematic with stage 2 and 3 omitted. Let us assume that the input voltage at IN+ changes by δV_{in} . Due to the high gain of the circuit, V_{out} can drive the output load with small changes to V_{BE} of Q1-4. Therefore, the change in voltage at emitter of Q2 (node a) or δV_a can be assumed to be 0 because IN- is at 0V.

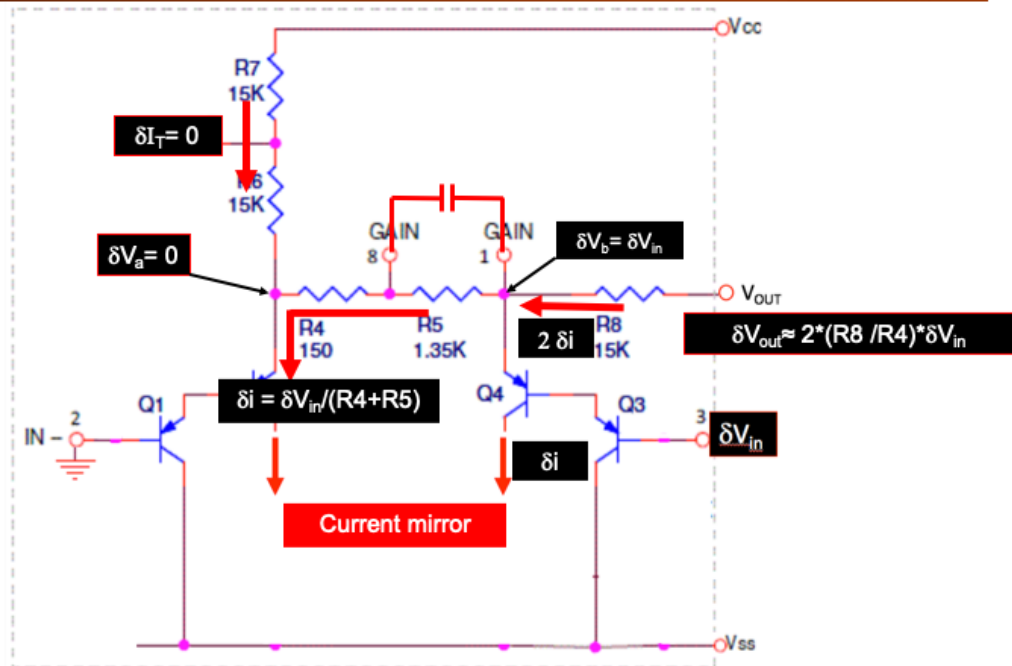
Similarly, δV_b at the emitter of Q4 (node b) can be assumed to be the same as δV_{in} . Hence the change in the current flowing from node b to a (i.e. through R4 and R5) is $\delta i = \delta V_{in} / (R4 + R5)$. This change in current is mirrored on the right branch of the differential pair through Q4. This current cannot come from the tail current because V_a has not changed and therefore $\delta I_T = 0$.

As a result, both δi through Q2 and Q4 must come from the R8 feedback resistor. Therefore δV_{in} results in $2\delta i$ flowing through R8.

Consequently, $\delta V_{out} = 2\delta i R8 + \delta V_{in} = 2 \cdot \delta V_{in} R8 / (R4 + R5) + \delta V_{in}$.

Assuming $\delta V_{out} \gg \delta V_{in}$, $\delta V_{out} / \delta V_{in} \approx 2 R8 / (R4 + R5)$.

LM386 Gain is $2 \cdot R8 / R4 \approx 200$ with bypass capacitor



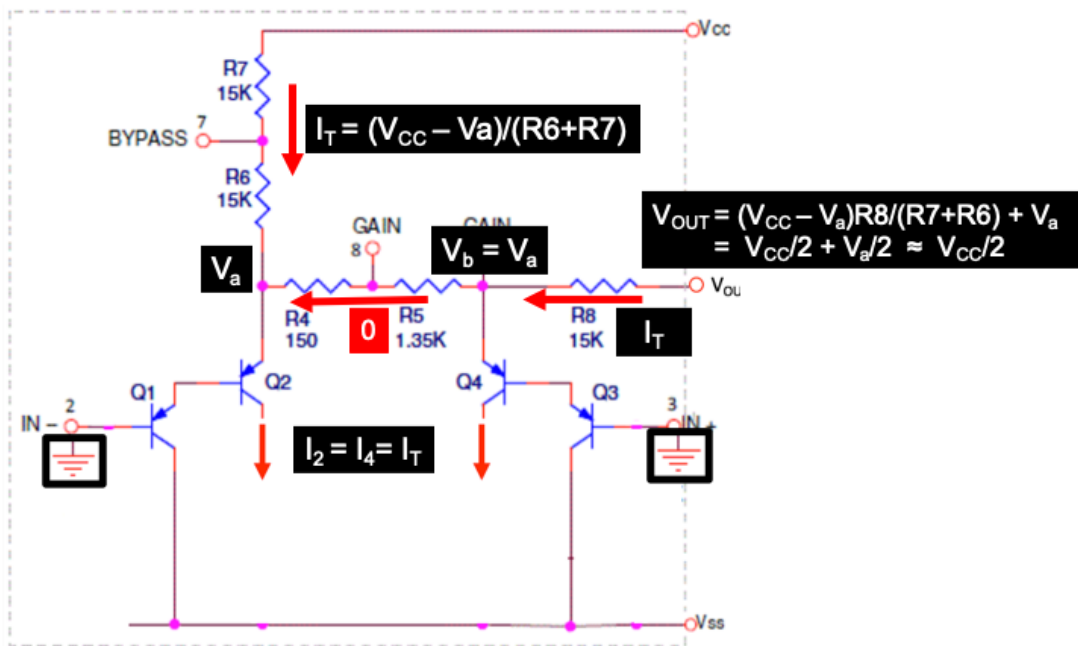
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LM386 provides access to bypass $R5$ with a 10uF capacitor. Then the gain of the amplifier increases to x 200.

LM386 Vout self biasing to $V_{CC}/2$



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The LM385 with the feedback network $R4$, $R5$ and $R8$ is automatically biasing its output V_{out} to approximately $V_{CC}/2$. This offset provides a maximum output voltage swing. This is how it works.

Assuming both inputs are zero (quiescent state). V_a and V_b must be the same because $Q2$ and $Q4$ are matched. No current flows through $R4$ and $R5$.

Therefore tail current I_T must flow through $Q2$. Current mirror reflects this current on $Q4$. Hence $I_2 = I_4 = I_T$. The same current I_T must flow from $R8$.

Hence, $V_{out} = I_T \cdot R8 + V_a$.

Since, $I_T = (V_{CC} - V_a)/(R6 + R7)$,

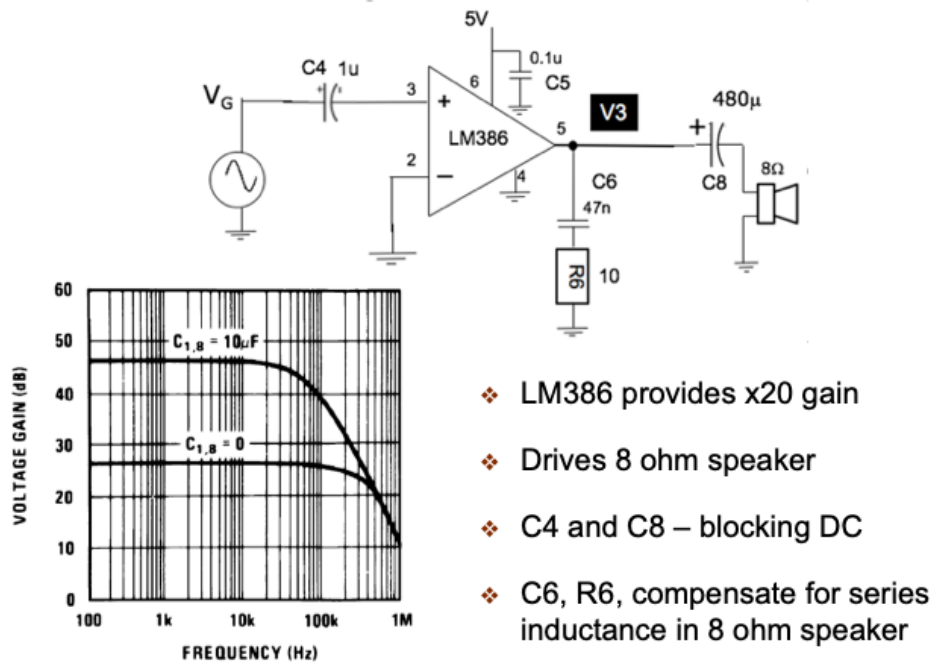
$$V_{out} = (V_{CC} - V_a) R8 / (R6 + R7) + V_a$$

Assuming that $V_a \ll V_{out} \ll V_{CC}$,

$$V_{out} \approx V_{CC}/2.$$

This is particularly true when V_{CC} is, say, 10V.

LM386 as a practical x20 amplifier (Lab 1)



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Lecture 3 Slide 21

In Lab 1, you will build a x20 amplifier driving an 8 Ω speaker.

C_4 and C_8 provide AC coupling for both input and output signals.

The input capacitor C_4 prevents the signal source from disturbing the DC bias for the LM386.

The output capacitor C_8 prevents the output offset of $V_{CC}/2$ being applied to the speaker, which may result in damage to the load.

The speaker is NOT equivalent to just an 8 Ω resistor. The voice coil of the speaker has an equivalent circuit of an inductor in series with a resistor. As a result, the load impedance changes with frequency, with the loading increases at higher frequencies. This can result in the phase of the output signal changes at higher frequencies in such a way that the internal feedback can cause oscillation to occur. C_6 and R_6 , the series RC network at the output provides compensation and avoid instability.